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Paleolatitude of glaciogenic upper Precambrian Rapitan Group and the use of tillites as chronostratigraphic marker horizons

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ABSTRACT

Currently available paleomagnetic evidence for the depositional paleolatitude of upper Precambrian glacial horizons is inconclusive. Unless it can be shown that glaciation occurred as a synchronous world-wide event, the use of tillities as chronostratigraphic marker horizons is not warranted. Paleomagnetic analysis of the upper Precambrian glaciogenic Lower Rapitan Subgroup in northwestern Canada has revealed three remanence directions. Remanence *X* is indicative of low paleolatitude, whereas remanence *Y* was acquired in high paleolatitudes. The preferred sequence, *X* is older than *Y*, is based on the evidence of stratigraphic considerations and prolonged chemical leaching. A third remanence, *Z*, is related to late deformation and (or) exhumation of the Rapitan Group.

INTRODUCTION

Two hypotheses have been proposed to explain the world-wide occurrence of upper Precambrian tillites (Crawford and Daily, 1971). The first considers these deposits to be diachronous and hence related to polar migration (McElhinny and others, 1974). The second hypothesis considers all the deposits to be synchronous and thus implies that glacial conditions occurred at low latitudes (Harland, 1964; Dunn and others, 1971; Kroner and Rankama, 1973). If synchronous, glacial beds would be global chronostratigraphic marker horizons and could be used to subdivide late Precambrian time (Harland, 1964).

Previous attempts to relate paleomagnetic data to upper Precambrian tillites have depended on the construction of generalized polar paths (based on few data) from which paleolatitudes have been deduced (Piper and others, 1973; McElhinny and others, 1974). The scarcity of paleomagnetic data and the possibility of large polar shifts during the interval spanned by the glacial beds mean that such paleolatitude interpolations are inconclusive (Bucha, 1965; McElhinny, 1973). For example, the late Precambrian polar path for Gondwanaland has been interpreted by some authors as confirming the first hypothesis (McElhinny and others, 1974), whereas others have held that the second hypothesis is correct (Piper, 1973; Piper and others, 1973). As has been shown for the late Paleozoic glaciation of Gondwanaland (Irving, 1966; McElhinny and Opdyke, 1968), arguments in favor of either hypothesis can be truly evaluated only after the paleolatitude of glacial horizons themselves (or of the strata that bound them on either side) has been reliably determined.

As an initial step toward an investigation of the upper Precambrian glacial horizons of North America, paleomagnetic data are here reported from the Rapitan Group in Canada. At its type locality in the Mackenzie Mountains, Yukon and Northwest Territories, the Rapitan Group has been subdivided into the Lower Rapitan Subgroup (400 m), dark purple laminated and bedded mudstone with interbeds of siltstone; Middle Rapitan Subgroup (500 m), polymictic conglomerate; and upper Rapitan Subgroup (600+ m), fissile shale with siltstone and minor limestone (Upitis, 1966; Young, 1976). One unusual aspect of the Rapitan Group is the widespread occurrence of a finely banded iron formation at the top of the Lower Rapitan (Gabrielse and others, 1973). In the study area the iron formation consists of 20 m of fine-bedded to laminated hematite and jasper containing as much as 40% iron. Geologically, the iron formation is problematical; it contains evidence of salt casts and also abundant evidence of glaciogenic deposition—for example, faceted and striated clasts, dropstones, and till pellets (Young, 1976).

On lithostratigraphic grounds the Rapitan has been correlated with similar extensive conglomerates (with or without iron formations) known from the Alaska-Yukon

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border in the north to the United States–Mexico border in the south (Brabb and Churkin, 1965; Crittenden and others, 1971; Stewart, 1972, 1976). These correlations do not prove that the tillites are synchronous; they merely indicate that the tillites have similar lithologic characteristics and exhibit similar relationships to the strata that bound them. At each locality, overlying Cambrian strata provide a rough upper age limit; there may be as much as 1,000 m of sedimentary rocks (commonly containing unconformities) between the Cambrian boundary and the top of the tillites. An older age limit is also poorly defined. Generally, the tillites lie unconformably on deformed Belt-Purcell strata. There are no radiometric ages available from Belt-Purcell strata in the Rapitan region. Radiometric data from the Canada–United States border region suggest sedimentation in the Belt area continued up to 1,100 m.y. B.P. and possibly up to 900 m.y. B.P. (Obradovich and Peterman, 1968). Deposition of the tillite, therefore, is bracketed in the interval 1,100 to 570 m.y. B.P. (that is, Hadrynian). One K-Ar study suggesting an age of between 825 and 900 m.y. has been reported from volcanic rocks (Miller and others, 1973) that directly overly the tillite in Washington State, but extensive secondary alteration of the lava flows means that this date is only tentative.

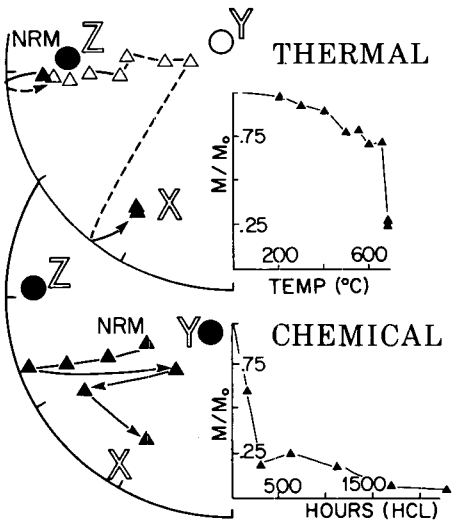


Figure 1. Above, magnetic direction and relative intensity changes of representative specimens during thermal demagnetization; equal-area projection. Below, example of changes of direction and relative intensity due to chemical leaching. NRM of each specimen may be resultant of a number of temporally distinct magnetizations. Large capital letters indicate vectors removed during cleaning. Open symbols conventionally indicate upward-directed remanence; solid symbols, downward-directed remanence.

RESULTS

Samples were collected from the type locality (lat 63.5°N, long 233.0°E) of the Rapitan Group in the Mackenzie Mountains, Northwest Territories. Seventeen sites (100 cores) were obtained from two parallel sections spaced 2 km apart in the Lower Rapitan Subgroup. In both sections, individual sites were occupied at measured intervals. Two other sites, one from the iron formation at the top of the Lower Rapitan and one from the mixtite base of the unconformably overlying Middle Rapitan, were also collected. Total stratigraphic thickness was in excess of 400 m, and the beds dipped uniformly to the southwest. The cores were sliced into 156 specimens and subjected to alternating field (a.f.), thermal, and chemical cleaning.

Chemical cleaning consisted of immersing the specimen in hydrochloric acid (10 *N*) for specified periods, washing in water, measuring and reimmersing in fresh acid. Acid penetration was aided by fine serration of the specimens. After the 2,140-h treatment, most of the red hematite cement had been removed. Thermal demagnetization on duplicate and other specimens was applied in 100 °C steps up to 500 °C and then in 25 °C steps up to 675 °C. A.f. demagnetization in 100-Oe steps up to 1,000 Oe produced little direction or intensity change. Most specimens carried more than one magnetization. To separate the individual remanence directions, vector subtraction was required (Fig. 1, chemical *Z* and *Y* directions). Separation of these directions was based on the assumption that for each cleaning interval, one particular remanence phase would be preferentially removed. Depending on the blocking-temperature spectrum (or the resistance of each remanence to leaching) recorded by a specimen, it is possible that the vector removed during a particular cleaning interval may actually represent the summation of two (or more) different directions (Roy and Park, 1974). Verification of a particular remanence was established by directly measuring it in other specimens of the same collection and (or) by its frequency in a large number of specimens. During cleaning of a single specimen, the same remanence direction may be removed during a number of individual cleaning intervals. Arbitrarily, where the removed vector directions from a single specimen did not differ by 20° between cleaning intervals, a vector mean of all removed vector directions was calculated, using the magnitude as a weighting measure. The resultant vector was taken as representative of one specimen remanence direction.

Application of this technique to both chemical and thermal cleaning data revealed three separate directions of magnetization, *X, Y, Z* (Fig. 2, Table 1). The *X* magnetization was preferentially revealed by prolonged chemical leaching (Fig. 2) and over a range of blocking temperatures (500 to 690 °C); in some instances, repeated treatment at 675 °C or treatment at 690 °C revealed this direction (Fig. 1). These magnetic properties are consistent with the *X* remanence originating in detrital grains of hematite. The broad spectrum of blocking temperatures probably arises from the variations of grain size and composition of the individual grains. The 690 °C direction, which is the same as that obtained from the thoroughly leached specimens, can be identified with coarse-grained hematite. This remanence must predate that of the

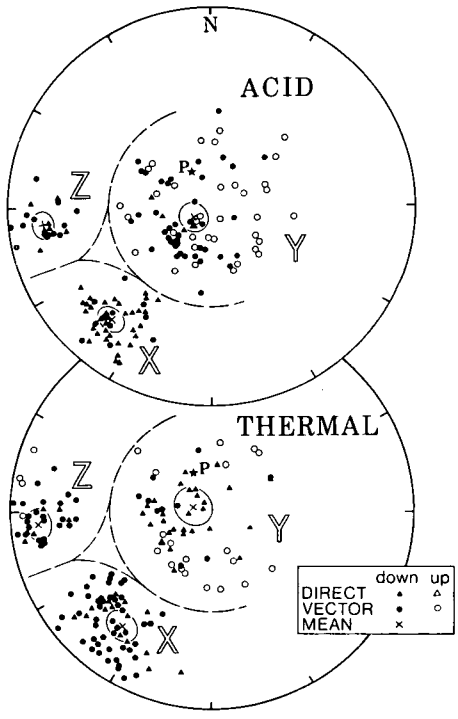


Figure 2. Equal-area projection of all direct measurement and vector-analysis remanence directions obtained by chemical leaching (above) and thermal demagnetization (below). All directions are given after correction for bedding tilt. In each case, directions apparently fall into three spatially separate fields (*X*, *Y*, and *Z*). *P*, which indicates present field direction at sampling locality, does not fall in error circle of either the thermal or chemical *Y* remanence mean.

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red cement since it is impossible (1) to reset coarse hematite thermally without resetting the normal *Y* remanence that blocks between 625 and 675 °C or (2) to reset the coarse hematite chemically without affecting the finer-grain cement fraction.

The normal *Y* remanence has blocking temperatures limited to a narrow interval (625 to 675 °C) just below the Curie point of hematite, which reflects the limited compositional and grain-size variation of its remanence-carrying phase. As shown by Figure 2, the *Y* remanence is rarely found by direct measurement after extirpation of the red pigment by acid leaching. Correlation of the two antiparallel *Y* directions between the two measured sections is not possible, since not only may specimens within a particular site record either polarity but some individual specimens record *both* polarities. Demagnetization of specimens with both polarities always removed the reversed *Y* direction before revealing the normal *Y* direction. As shown in Figure 2 the reversed *Y* direction is never found by direct measurement after extensive thermal or chemical treatment. This suggests a gradual acquisition of the *Y* direction during a period of change from normal to reversed polarity (compare Roy and Park's [1974] study of Carboniferous red beds). *Z*, the least commonly found magnetization, rarely exhibits high blocking temperatures (<500 °C), has low resistance to acid leaching, and is occasionally revealed by a.f. cleaning.

All three directions *X*, *Y*, and *Z* are found in samples of the Lower Rapitan Subgroup. Specimens from the iron formation and the basal mixtite of the Middle Rapitan exhibit only the *Y* and *Z* magnetizations.

DISCUSSION

The *X* and *Y* directions indicate totally different depositional paleolatitudes—equatorial if *X* is the depositional magnetization and polar if *Y* is valid. Since the *X* direction is restricted to the stratigraphically older Lower Rapitan (where it is most commonly found after prolonged acid leaching) whereas the *Y* direction appears to have been slowly acquired during a change from normal to reversed polarity, the preferred magnetization sequence is $X \rightarrow Y$. *Z* is interpreted in accord with its low magnetic stability as due to a thermal overprint. The scarcity of late Precambrian paleomagnetic data (Fig. 3) makes a number of interpretations of the Rapitan results possible. On the basis of pole paths drawn up from other late Precambrian data, the simplest interpretation is that pole *X* represents the original depositional remanence of the Lower Rapitan which formed in low latitudes at approximately 1,000 m.y. B.P. and pole *Y* represents a diagenetic overprint recorded by red hematite pigment which appears to be associated with deposition of the massive conglomerates (tillites) of the Middle Rapitan in high latitudes at approximately 960 m.y. B.P.

Other more complex interpretations of the data are possible. This study has been very detailed, yet it is still not possible to evaluate conclusively the depositional latitude of these tillites. In the above interpretation it has been assumed that *Y* remanence dates from late Precambrian time, since this is most compatible with the available paleomagnetic and geologic evidence. Although it seems unlikely that the red hematite pigment was not acquired until approximately 900 m.y. after initial deposition of the Rapitan Group, the similarity between the *Y* direction and the present Earth's field could also be interpreted as evidence of a recent magnetization. This highlights a paleomagnetic dilemma in defining the paleolatitude of all tillites. Proving that a steeply inclined remanence is of recent or ancient origin is of equal importance, and instead of using recent remagnetization as a convenient sack, it is more correct to consider *all* possibilities consistent with the available field and laboratory evidence. For example, two older studies of upper Precambrian tillite from east Greenland and Norway (Harland and Bidgood, 1959; Bidgood and Harland, 1961) each yielded two directions, one of steep inclination and one with shallow inclination; no cleaning of the remanence was attempted, and the relative stability of the various directions was not established. The steeply inclined direction was conveniently attributed to recent remagnetization. Having then to rely on the shallowly inclined direction (poles SP, EG, Fig. 3) these workers stated, perhaps erroneously, that the paleomagnetic evidence favored low-latitude glaciation. More recently, paleomagnetic analysis of the Port Askaig Tillite of Scotland has also yielded two remanence directions of known stability but unknown age, one steeply inclined and one shallowly inclined (Tarling, 1974). Again the ambiguity of high- or low-latitude glaciation was not resolved. The sequence of remanence acquisition was interpreted solely from a comparison with the currently deficient early Paleozoic and late Precambrian paleomagnetic record. Furthermore, since the Port Askaig Til-

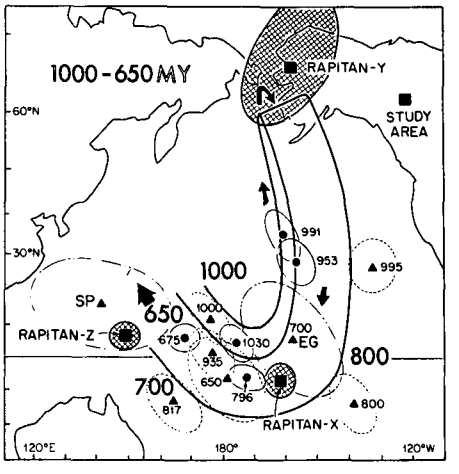


Figure 3. Latest Precambrian (650 to 1,000 m.y. B.P.) poles from Laurentia and possible polar path modified after Irving and McGlynn (1976). Rapitan-*X*, Rapitan-*Y*, and Rapitan-*Z* are the three poles from Rapitan Group. Preliminary poles from upper Precambrian tillites from east Greenland (EG) and southern Norway (SP) have been plotted after correction for Mesozoic opening of Atlantic. Poles of individual rock units are indicated by their ages in millions of years, as follows: 650, Jacobsville Sandstone; 675, Frankling Diabase; 796, Torridon Group; 800, Multicoloured series; 817, Frontenac Dykes; 935, Rama Diabase; 953, El Paso rocks; 991, Stoer Group; 995, Aillik Dykes; 1,000, Nankoweap Formation; and 1,030, Pike's Peak Granite. Results that conform to the A (age ± 50 m.y.) category of age reliability (Irving and McGlynn, 1976) are represented by dots; poles falling into the B (age ± 100 m.y.) and C (age ± 200 m.y.) categories of age reliability are represented by triangles.

TABLE 1. THREE DIRECTIONS OF MAGNETIZATION

	*N**	*D*†	*I*†	*R*†	*k*†	α_{95}†	Lat	Long
X (tilt corrected)	19	220	26	18.66	54	5	07°S	194°E
Y (tilt corrected)	34	294	82	32.09	17	6	66°N	197°E
Z (in situ)	17	267	15	16.79	76	4	05°N	149°E

*Because of the complex remanence characteristics of these specimens, the following hierarchical system was adopted. Where demagnetization revealed the same removed vector a number of times, a vector mean was calculated to give specimen direction means. Each removed vector was weighted with respect to its magnitude. Site means were calculated by giving unit weight to each specimen mean, irrespective of whether that mean direction was found by chemical or thermal cleaning. Hence *N* in the above calculations represents the number of sites at which a particular component has been identified. In the case of the *Y* component, this represents the mean, irrespective of polarity of both the −*Y* and +*Y* components.

†*D* = mean declination; *I* = mean inclination; *R*, *k*, and α_{95} are standard Fisher (1953) paleomagnetic parameters.

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lite at the sampling locality exhibited two cleavages, it seems questionable whether the primary remanence could ever be recovered.

Finally, because late Precambrian polar paths are so poorly defined and because steeply inclined directions have been ignored, it is not correct to interpolate the paleolatitude of tillites. More data are required from widely separated tillite occurrences before it is possible to decide whether glaciation took place in high or low latitudes. As yet, the use of upper Precambrian tillites as *chronostratigraphic* marker horizons—on the assumption that they result from some catastrophic, synchronous, world-wide glaciation—is not warranted.

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